Generalizing Diagonalization

Recall that we can only diagonalize square matrices, but we can do something close! But, we also really like orthogonal matrices, so if we can also do that, it would make our lives really easy! In other words, let's try to find 3 matrices such that for *any* $A \in \mathbb{R}^{m\times n}$:

$$A = U\Sigma V^\top$$

Where $U \in \mathbb{R}^{m\times m}, \Sigma \in \mathbb{R}^{m\times n}, V \in \mathbb{R}^{n\times n}$ such that

$$U^{-1} = U^\top \qquad V^{-1} = V^\top$$

and

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & \dots & \dots & \dots & 0 \\ 0 & \sigma_2 & \ddots & \dots & \dots & \dots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \dots & \dots & \vdots \\ \vdots & \vdots & \ddots & \sigma_r & \ddots & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 0 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & \dots & \dots & 0 \end{bmatrix} = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}_{m\times n} \qquad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$

Motivating Our Method

Let's assume that we can find such matrices. This would mean that

$$A^\top A = V\Sigma^\top \Sigma V^\top \qquad AA^\top = U\Sigma\Sigma^\top U^\top$$

Where

$$\Sigma^\top\Sigma = \begin{bmatrix} \Sigma_r^2 & 0 \\ 0 & 0 \end{bmatrix}_{n\times n} \qquad \Sigma\Sigma^\top = \begin{bmatrix} \Sigma_r^2 & 0 \\ 0 & 0 \end{bmatrix}_{m\times m}$$

$A^\top A$ and $AA^\top$ are Orthogonally Diagonalizable

Theorem

For any $A \in \mathbb{R}^{m \times n}$, both $A^\top A$ and $AA^\top$ are orthogonally diagonalizable.

Proof.

Recall that the Real Spectral Theorem states that a real matrix is orthogonally diagonalizable if it is symmetric. See that

$$\left(A^\top A\right)^\top = A^\top \left(A^\top\right)^\top = A^\top A$$

$$\left(AA^\top\right)^\top = \left(A^\top\right)^\top A^\top = AA^\top$$

$\square$

All Eigenvalues of $A^\top A$ are non-negative

Theorem

If λ is an eigenvalue of $A^\top A$, then it is non-negative.

Proof.

Let $(\lambda, \mathbf{v})$ be an eigenpair of $A^\top A$. See that

$$\|A\mathbf{v}\|_2^2 = (A\mathbf{v})^\top A\mathbf{v} = \mathbf{v}^\top A^\top A\mathbf{v} = \mathbf{v}^\top \left(A^\top A\mathbf{v}\right) = \lambda \mathbf{v}^\top \mathbf{v} = \lambda \|v\|_2^2$$

Rearranging above and remembering that $\|\mathbf{v}\|_2 > 0$, we get that

$$\lambda = \frac{\|A\mathbf{v}\|_2^2}{\|\mathbf{v}\|_2^2} \geq 0$$

$\square$

$\mathrm{Nul}\,(A) = \mathrm{Nul}\,(A^\top A)$

Theorem

If $A \in \mathbb{R}^{m\times n}$, then $\mathrm{Nul}\,(A) = \mathrm{Nul}\,(A^\top A)$.

Proof.

First, we show that $\mathrm{Nul}\,(A) \subseteq \mathrm{Nul}\,(A^\top A)$.
Let $\mathbf{v} \in \mathrm{Nul}\,(A)$. This means $A\mathbf{v} = \mathbf{0}_m$. See that

$$A^\top A\mathbf{v} = A^\top \mathbf{0}_m = \mathbf{0}_n$$

Now, we show that $\mathrm{Nul}\,(A^\top A) \subseteq \mathrm{Nul}\,(A)$.
Let $\mathbf{v} \in \mathrm{Nul}\,(A^\top A)$. This means $A^\top A\mathbf{v} = \mathbf{0}_n$. So, $A\mathbf{v} \in \mathrm{Nul}\,(A^\top)$ and $A\mathbf{v} \in \mathrm{Col}\,(A)$. But, the Fundamental Theorem of Linear Algebra states that $\mathrm{Col}\,(A)^\perp = \mathrm{Nul}\,(A^\top)$. This means that $A\mathbf{v} = \mathbf{0}_m$, so $\mathbf{v} \in \mathrm{Nul}\,(A)$ as desired. $\square$

A consequence is $\mathrm{rank}\,(A) = \mathrm{rank}\,(A^\top A)$ via the rank-nullity theorem.

The SVD

Theorem

If $A \in \mathbb{R}^{m\times n}$ and $\operatorname{rank}(A) = r \leq \min(m,n)$, then there exists some $U \in \mathbb{R}^{m\times m}, \Sigma \in \mathbb{R}^{m\times n}, V \in \mathbb{R}^{n\times n}$ such that

$$A = U\Sigma V^\top = \begin{bmatrix} \mathbf{u}_1 & \dots & \mathbf{u}_r & \mathbf{u}_{r+1} & \dots & \mathbf{u}_m \end{bmatrix} \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}_{m\times n} \begin{bmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_r^\top \\ \mathbf{v}_{r+1}^\top \\ \vdots \\ \mathbf{v}_n^\top \end{bmatrix}$$

and

$$U^{-1} = U^\top, V^{-1} = V^\top, \Sigma_r = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

This is called the "singular value decomposition"

Proof of the Existence of the SVD Part 1

Since $A^\top A$ is orthogonally diagonalizable, let $A^\top A = VDV^\top$ be an eigendecompositon of $A^\top A$. Also, we sort the eigenvalues such that $\lambda_1 \geq \cdots \geq \lambda_r > \lambda_{r+1} = \cdots = \lambda_n = 0$ (repeated eigenvalues are given different numbers here). For $i = 1, \ldots, r$, define $\sigma_i = \sqrt{\lambda_i}$ and

$$\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$$

Next, pick $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_n$ to be an orthonormal basis of $\mathrm{Nul}\left(A^\top\right) = \mathrm{Col}\left(A\right)^\perp$.

Proof of the Existence of the SVD Part 2

Define $U = \begin{bmatrix} \mathbf{u}_1 & \dots & \mathbf{u}_m \end{bmatrix}, V = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_n \end{bmatrix}, \Sigma_r = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$, and $\Sigma = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}_{m \times n}$

We will now show that $AV = U\Sigma$

$$AV = \begin{bmatrix} A\mathbf{v}_1 & \dots & A\mathbf{v}_r & A\mathbf{v}_{r+1} & \dots & A\mathbf{v}_n \end{bmatrix} = \begin{bmatrix} \sigma_1\mathbf{u}_1 & \dots & \sigma_r\mathbf{u}_r & \mathbf{0}_m & \dots \mathbf{0}_m \end{bmatrix}$$

$$U\Sigma = \begin{bmatrix} \mathbf{u}_1 & \dots & \mathbf{u}_r & \mathbf{u}_{r+1} & \dots & \mathbf{u}_m \end{bmatrix} \Sigma = \mathbf{u}_1\mathbf{e}_1^\top\sigma_1 + \dots + \mathbf{u}_r\mathbf{e}_r^\top\sigma_r + \mathbf{u}_{r+1}\mathbf{0}_n^\top + \dots + \mathbf{u}_m\mathbf{0}_n^\top$$

$$= \begin{bmatrix} \sigma_1\mathbf{u}_1 & \dots & \sigma_r\mathbf{u}_r & \mathbf{0}_m & \dots \mathbf{0}_m \end{bmatrix} = AV$$

Thus, we have that

$$AV = U\Sigma \to A = U\Sigma V^\top$$

Computing the SVD

Our proof is called a "constructive" proof because it gives us a way to actually compute it! Thus, an algorithm for computing the SVD is:

1. Compute the eigenvectors and eigenvalues of $A^\top A$, sort them in descending order based on the eigenvalues. Perform Gram-Schmidt if necessary!
2. Compute $\sigma_1 = \sqrt{\lambda_1}, \ldots, \sigma_r = \sqrt{\lambda_r}$
3. Compute $\mathbf{u}_1 = \frac{1}{\sigma_1} A\mathbf{v}_1, \ldots, \mathbf{u}_r = \frac{1}{\sigma_r} A\mathbf{v}_r$
4. Find an orthonormal basis of $\mathrm{Nul}\left(A^\top\right)$, and call it $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$.

Computing the SVD Example Part 1

Let's find the SVD of $A = \begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix}$.

$$A^\top A = \begin{bmatrix} 333 & 81 \\ 81 & 117 \end{bmatrix}$$

This matrix has eigenvalues $\lambda_1 = 360, \lambda_2 = 90$ with $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$ being their associated eigenvectors.

So,

$$V = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sqrt{360} & 0 \\ 0 & \sqrt{90} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 6\sqrt{10} & 0 \\ 0 & 3\sqrt{10} \\ 0 & 0 \end{bmatrix}$$

Computing the SVD Example Part 2

Now, we compute $\mathbf{u}_1, \mathbf{u}_2$.

$$\mathbf{u}_1 = \frac{1}{\sigma_1} A\mathbf{v}_1 = \frac{1}{60}\begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \frac{1}{60}\begin{bmatrix} 20 \\ 40 \\ 40 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\mathbf{u}_2 = \frac{1}{\sigma_2} A\mathbf{v}_2 = \frac{1}{30}\begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix}\begin{bmatrix} 1 \\ -3 \end{bmatrix} = \frac{1}{30}\begin{bmatrix} -20 \\ -10 \\ 20 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$$

Computing the SVD Example Part 3

Now, to compute $\mathbf{u}_3$, we need to find an orthonormal basis of $\mathrm{Nul}\left(A^\top\right)$.

$$\begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} \xrightarrow{R_2 = R_2 - 2R_1} \begin{bmatrix} 4 & 11 & 14 \\ 0 & -15 & -30 \end{bmatrix} \xrightarrow{R_2 = -\frac{1}{15}} \begin{bmatrix} 4 & 11 & 14 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_1 = R_1 - 11R_2} \begin{bmatrix} 4 & 0 & -8 \\ 0 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow{R_1 = \frac{1}{4}R_1} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \end{bmatrix}$$

So, a basis of $\mathrm{Nul}\left(A^\top\right)$ is $\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$. In order for this basis to be orthonormal, we just divide it by it's norm to get $\mathbf{u}_3 = \frac{1}{3}\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$, so $U = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$

Computing the SVD Part 4

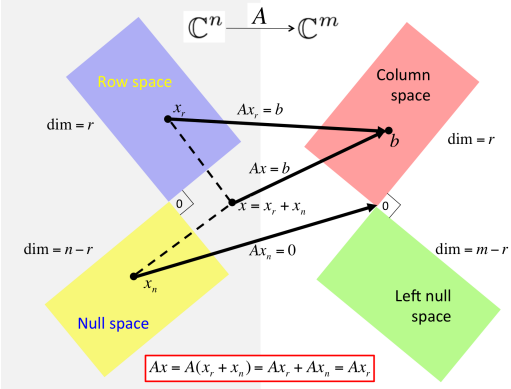
Thus, our SVD is

$$\begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}\begin{bmatrix} 6\sqrt{10} & 0 \\ 0 & 3\sqrt{10} \\ 0 & 0 \end{bmatrix}\frac{1}{\sqrt{10}}\begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix}$$

Relating the SVD to our Fundamental Subspaces

The following statements hold for any $A \in \mathbb{R}^{m \times n}$.

1. $\text{Row}(A) = \text{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_r)$
2. $\text{Nul}(A) = \text{Span}(\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n)$
3. $\text{Col}(A) = \text{Span}(\mathbf{u}_1, \ldots, \mathbf{u}_r)$
4. $\text{Nul}(A^\top) = \text{Span}(\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m)$



Source[a]

[a] https://www.cs.utexas.edu/~flame/laff/alaff/chapter04-four-fundamental-spaces.html